BIG MART SALES ANALYSIS

Retail Sales Analysis and Outlet Sales Prediction

Project Report

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1. Executive Summary

This project performs an end-to-end analysis of the Big Mart Sales dataset to understand retail sales performance across outlets, outlet types, product categories and location tiers. The work combines data cleaning, exploratory data analysis, visualisation, and predictive modelling to explain and predict Item_Outlet_Sales.

The notebook contains 8,523 observations and 12 variables. Missing Item_Weight values were handled using a Bayesian model implemented with PyMC, while missing Outlet_Size values were predicted using a Random Forest classifier. After preprocessing, the cleaned dataset contained no missing values.

For prediction, a Random Forest Regressor achieved a 5-fold cross-validated R^2 of 0.5578. XGBRFRegressor improved this to 0.5963, and the reduced-feature XGBRF model achieved 0.5964. On the final 20% test set, the model achieved MAE=714.73, RMSE=1019.73, and R^2=0.6174.

2. Project Objectives

- Analyze sales performance by outlet.
- Compare sales across outlet types and outlet location tiers.
- Evaluate item-type performance and identify high-performing product categories.
- Identify the top products by aggregated sales.
- Compare regional performance using total sales, average sales and outlet counts.
- Examine item-type sales patterns across individual outlets.
- Build machine-learning models to predict Item_Outlet_Sales.
- Identify the most influential predictors of outlet sales.

3. Dataset Description

The notebook loads a dataset with 8,523 rows and 12 columns. The schema corresponds to the Big Mart Sales problem and includes product, outlet and sales variables.

Variable	Description
Item_Identifier	Unique product identifier
Item_Weight	Weight of the product
Item_Fat_Content	Fat-content category
Item_Visibility	Product visibility in the outlet
Item_Type	Product category/type
Item_MRP	Maximum Retail Price
Outlet_Identifier	Unique outlet identifier
Outlet_Establishment_Year	Outlet establishment year
Outlet_Size	Outlet size category
Outlet_Location_Type	Tier-based location category
Outlet_Type	Outlet format/type
Item_Outlet_Sales	Item sales at the outlet; prediction target

4. Data Quality and Preprocessing

Initial inspection showed 1,463 missing Item_Weight observations and 2,410 missing Outlet_Size observations. The notebook addressed these issues before analysis.

4.1 Bayesian Imputation of Item Weight

A PyMC probabilistic model was used for Item_Weight. A Normal prior was specified for the mean and a Half-Normal prior for the standard deviation. Observed item weights were used in the likelihood, posterior samples were generated, and the posterior mean of the missing-value distribution was used to impute missing weights.

4.2 Random Forest Imputation of Outlet Size

Outlet_Type and Outlet_Location_Type were encoded and used to train a Random Forest classifier on rows where Outlet_Size was known. The trained classifier was then used to predict the missing Outlet_Size values.

4.3 Category Standardization

In Item_Fat_Content, LF and low fat were standardized to Low Fat, while reg was standardized to Regular.

After these preprocessing steps, the notebook reports zero missing values in all variables.

5. Exploratory Data Analysis

The sales distribution is right-skewed. Most observations are concentrated at low-to-moderate sales levels, while a smaller number of products have substantially higher sales.

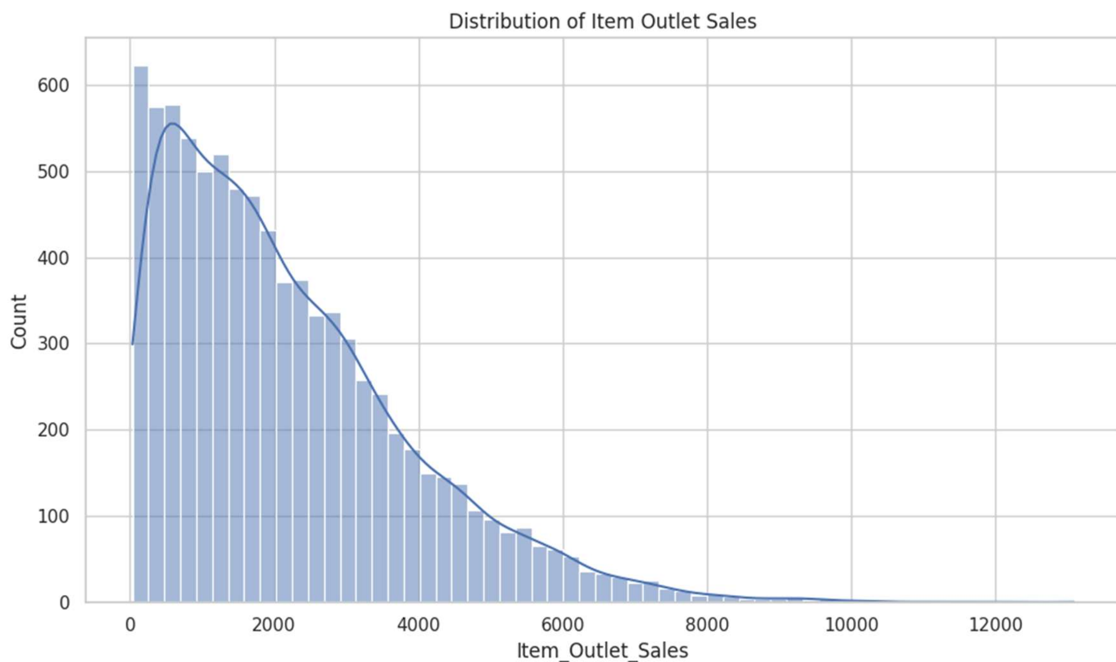

Figure 1. Distribution of Item Outlet Sales.

5.1 Sales Performance by Outlet

Outlet	Total Sales
OUT010	₹188,340
OUT013	₹2,142,664
OUT017	₹2,167,465
OUT018	₹1,851,823
OUT019	₹179,694
OUT027	₹3,453,926
OUT035	₹2,268,123
OUT045	₹2,036,725

OUT046	₹2,118,395
OUT049	₹2,183,970

OUT027 recorded the highest total sales at approximately ₹34.54 lakh. OUT019 and OUT010 recorded the lowest totals, at approximately ₹1.80 lakh and ₹1.88 lakh respectively.

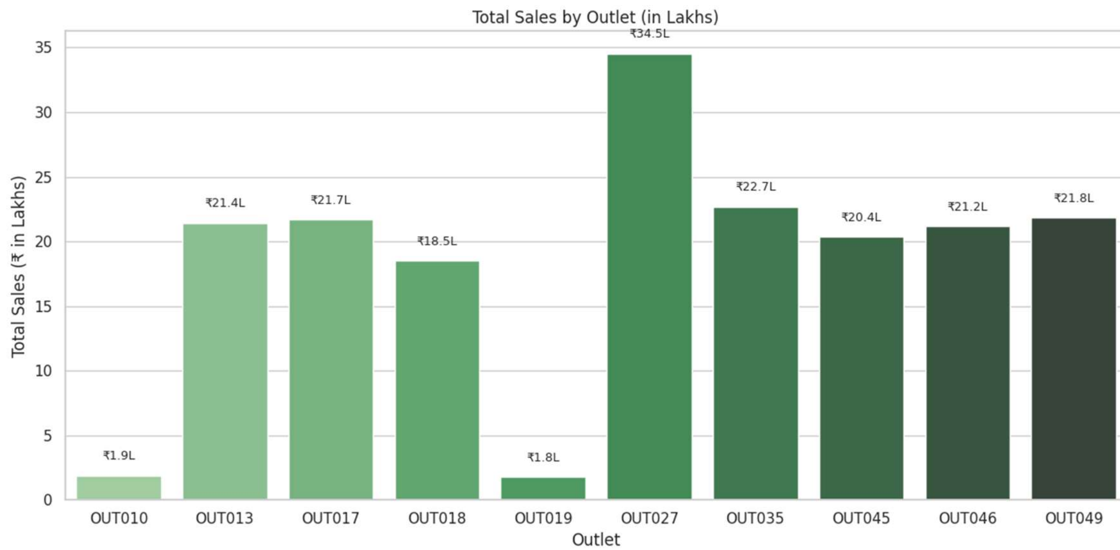

Figure 2. Total sales by outlet.

5.2 Sales by Outlet Type

The analysis shows substantial differences between outlet formats. Supermarket Type3 has the highest average sales among the outlet-type combinations shown, with mean item sales of approximately ₹3,694.

Outlet Type	Location	Average Sales	Total Sales
Grocery Store	Tier 1	₹340.33	₹1.80 lakh
Grocery Store	Tier 3	₹339.35	₹1.88 lakh
Supermarket Type1	Tier 1	₹2,313.10	₹43.02 lakh
Supermarket Type1	Tier 2	₹2,323.99	₹64.72 lakh
Supermarket Type1	Tier 3	₹2,299.00	₹21.43 lakh
Supermarket Type2	Tier 3	₹1,995.50	₹18.52 lakh
Supermarket Type3	Tier 3	₹3,694.04	₹34.54 lakh

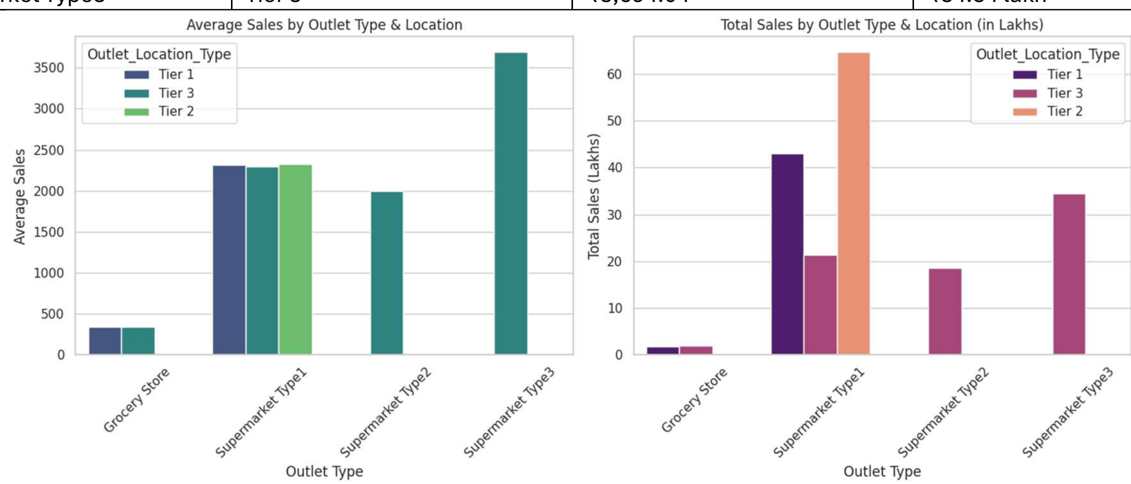

Figure 3. Sales by outlet type and location tier.

5.3 Item Type Performance

Rank	Item Type	Total Sales
1	Fruits and Vegetables	₹28.20 lakh
2	Snack Foods	₹27.33 lakh
3	Household	₹20.55 lakh
4	Frozen Foods	₹18.26 lakh
5	Dairy	₹15.23 lakh

Fruits and Vegetables is the highest-selling category, followed by Snack Foods. These categories therefore represent important contributors to aggregate retail sales.

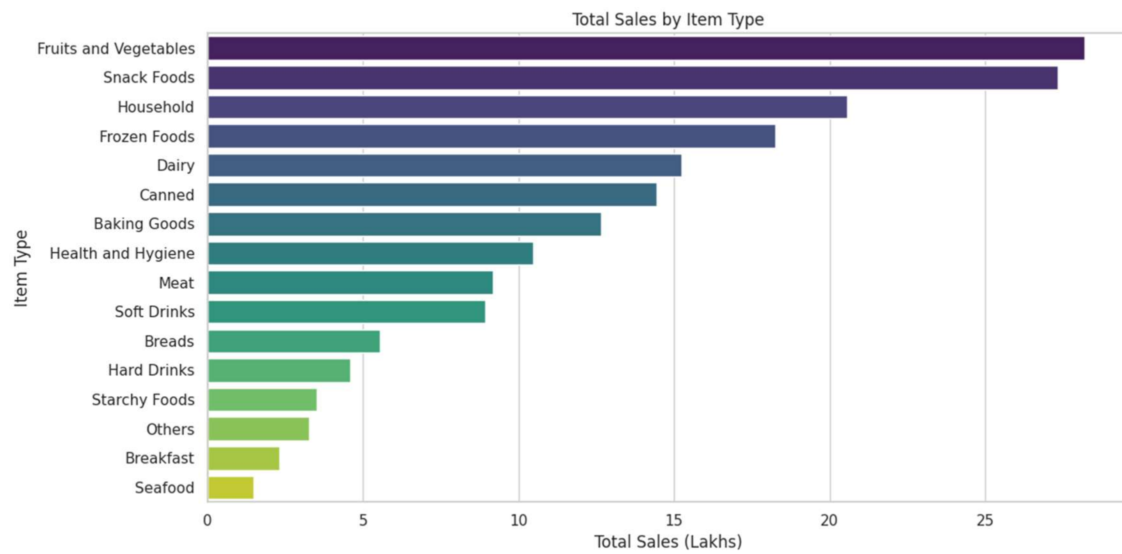

Figure 4. Total sales by item type.

5.4 Top Products by Total Sales

Rank	Product	Item Type	Total Sales
1	FDY55	Fruits and Vegetables	₹42,661.80
2	FDA15	Dairy	₹41,584.54
3	FDZ20	Fruits and Vegetables	₹40,185.02
4	FDF05	Frozen Foods	₹36,555.75
5	FDA04	Frozen Foods	₹35,741.48
6	FDK03	Dairy	₹34,843.98
7	NCQ06	Household	₹34,680.19
8	NCQ53	Health and Hygiene	₹34,508.41
9	FDJ55	Meat	₹33,531.02
10	FDD44	Fruits and Vegetables	₹32,723.40

FDY55 (Fruits and Vegetables) is the top individual product in the notebook's aggregated ranking, with total sales of approximately ₹42,661.80.

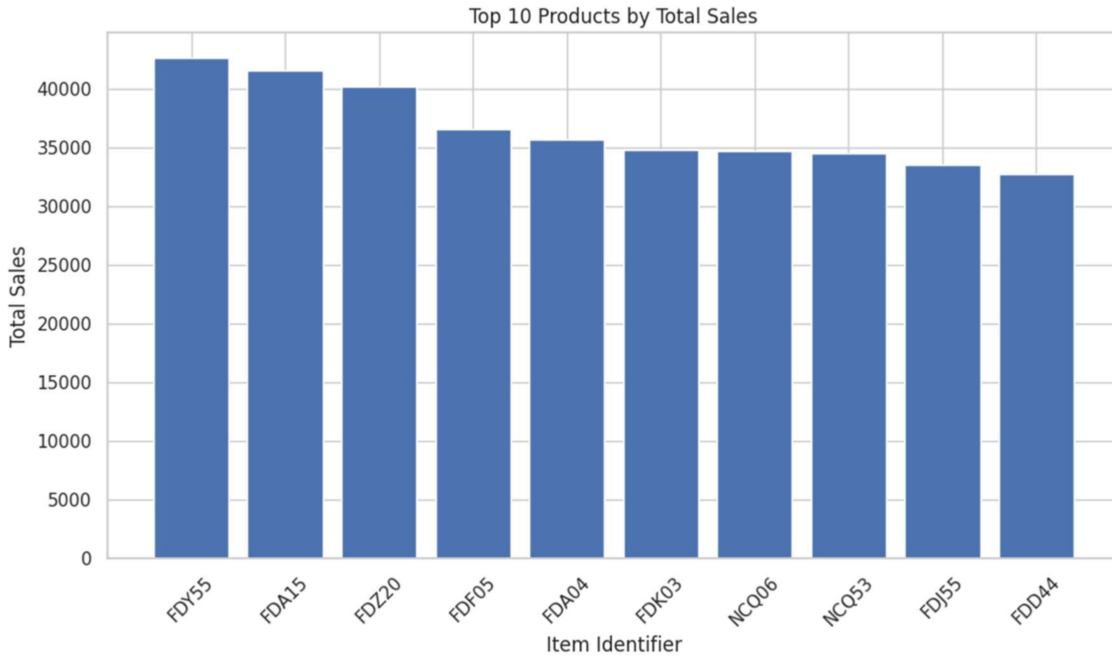

Figure 5. Top 10 products by total sales.

6. Regional Performance Comparison

Location Tier	Total Sales	Average Sales	Unique Outlets
Tier 3	₹76.37 lakh	₹2,279.63	4
Tier 2	₹64.72 lakh	₹2,323.99	3
Tier 1	₹44.82 lakh	₹1,876.91	3

Tier 3 generated the highest total sales, while Tier 2 recorded the highest average sales per item. Tier 3 also contained four unique outlets, compared with three each in Tier 1 and Tier 2.

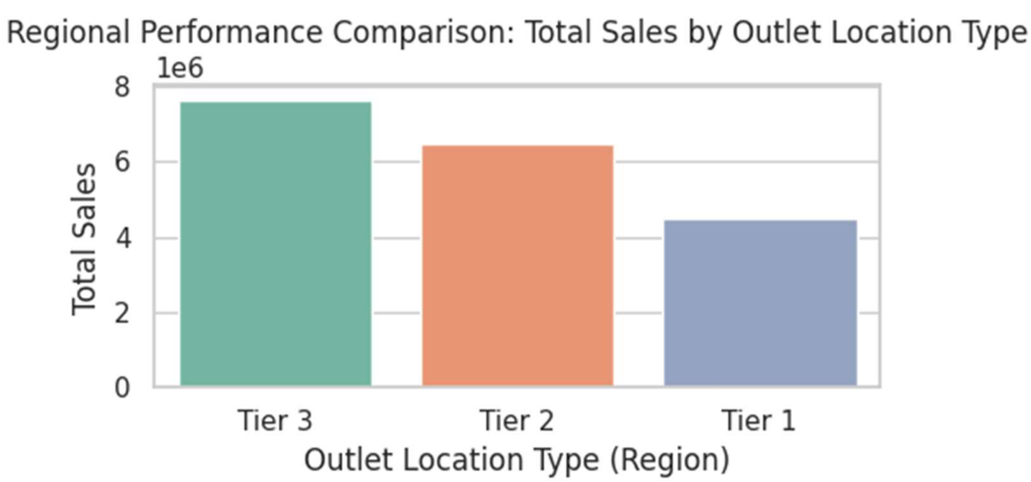

Figure 6. Total sales by outlet location tier.

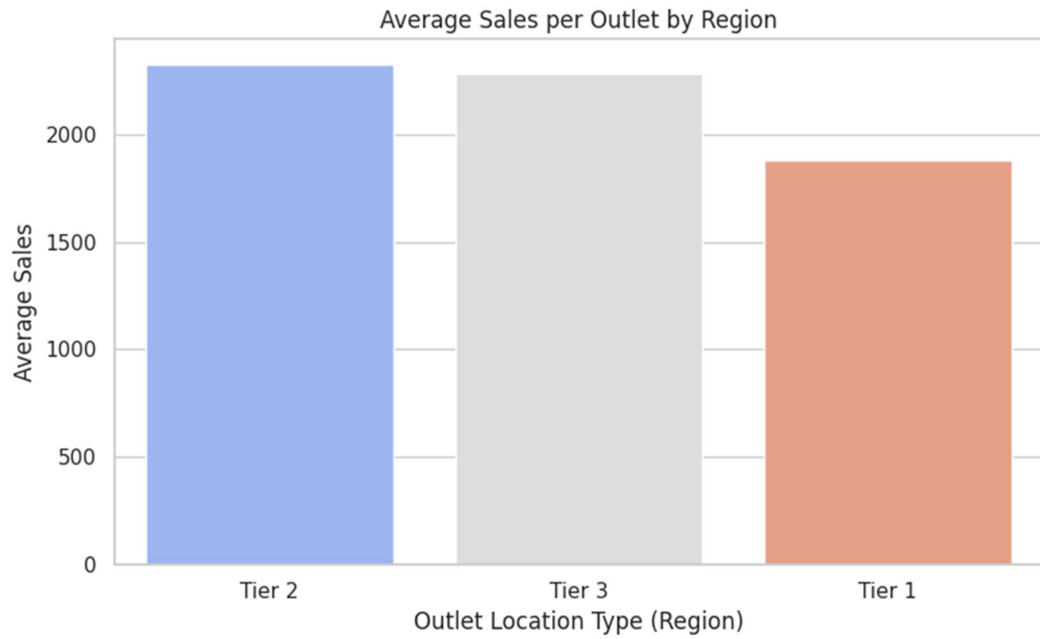

Figure 7. Average sales by outlet location tier.

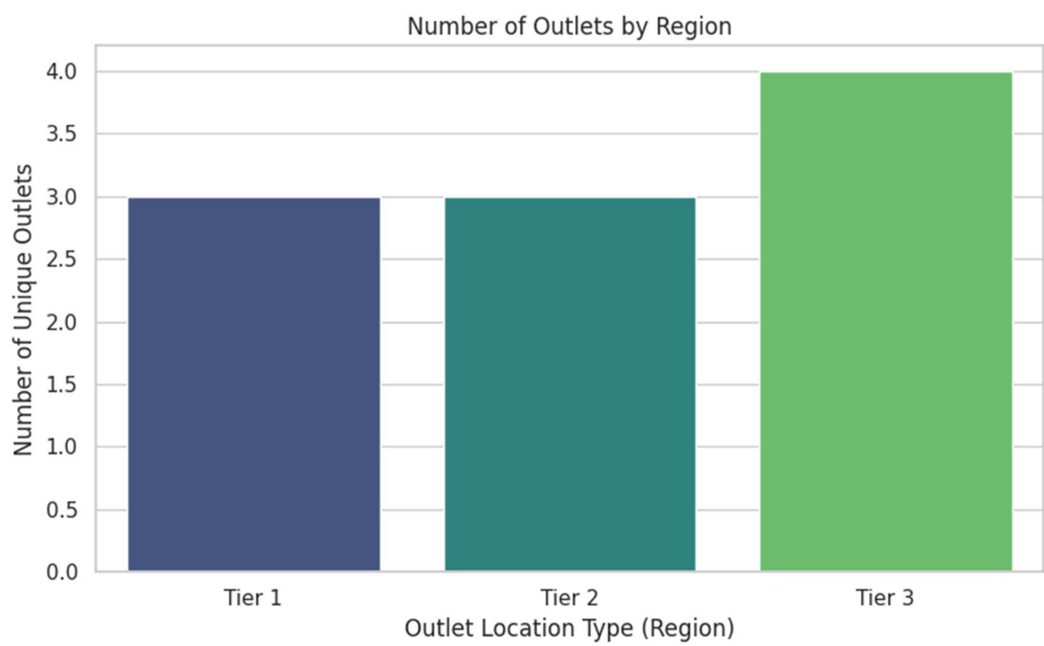

Figure 8. Number of unique outlets by location tier.

7. Item-Type Trends Across Outlets

The notebook compares item-type sales across individual outlets using line charts and heatmaps. The top-selling category differs across outlets, although Fruits and Vegetables and Snack Foods appear repeatedly among the strongest categories.

Outlet	Top Item Type	Sales
OUT010	Snack Foods	₹25,942.90
OUT013	Fruits and Vegetables	₹341,526.77
OUT017	Fruits and Vegetables	₹319,504.10

OUT018	Snack Foods	₹278,714.53
OUT019	Snack Foods	₹25,653.27
OUT027	Fruits and Vegetables	₹576,028.19
OUT035	Snack Foods	₹355,573.82
OUT045	Fruits and Vegetables	₹326,414.44
OUT046	Fruits and Vegetables	₹304,751.31
OUT049	Snack Foods	₹350,244.09

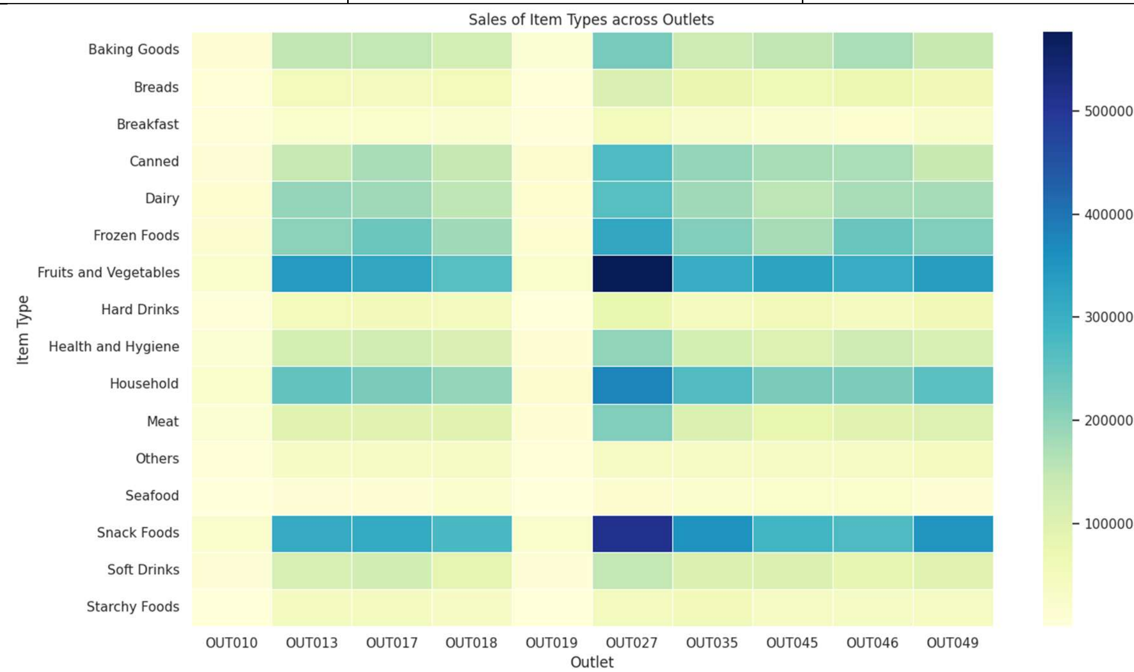

Figure 9. Heatmap of item-type sales across outlets.

8. Predictive Modelling

The target variable is Item_Outlet_Sales. Before modelling, Store_age was engineered as 2013 minus Outlet_Establishment_Year. Item_Identifier was reduced to its first two characters, and categorical variables were ordinal-encoded.

8.1 Model Comparison

Model	Validation Design	R^2
Random Forest Regressor	5-fold cross-validation	0.5578
XGBRFRegressor	5-fold cross-validation	0.5963
Reduced-feature XGBRFRegressor	5-fold cross-validation	0.5964

XGBRFRegressor outperformed Random Forest in cross-validation. Removing the lower-importance variables listed in the notebook resulted in a very small further improvement in mean R^2.

8.2 Feature Importance

Rank	Feature	Importance
1	Outlet_Type	0.440799
2	Store_age	0.173503
3	Item_MRP	0.164336
4	Outlet_Identifier	0.152056
5	Outlet_Location_Type	0.033122
6	Outlet_Size	0.025899
7	Item_Weight	0.003410
8	Item_Visibility	0.003083
9	Item_Type	0.002128

10	Item_Identifier	0.001100
11	Item_Fat_Content	0.000565

Outlet_Type is by far the most important predictor in the XGBRF model, followed by Store_age, Item_MRP and Outlet_Identifier. The remaining features have much smaller reported importance.

8.3 Final Test-Set Performance

Metric	Result
MAE	714.73
RMSE	1019.73
R^2	0.6174

Using an 80/20 train-test split, the final XGBRF model achieved an R^2 of approximately 0.6174. This indicates that the model explains about 61.7% of the observed variation in the test-set target under the notebook's modelling setup.

9. Sample Prediction

The notebook demonstrates a sample prediction using a manually supplied feature vector. The resulting predicted Item_Outlet_Sales value is approximately ₹727.99.

10. Key Findings

- OUT027 is the highest-sales outlet in the notebook's outlet-level aggregation.
- Fruits and Vegetables is the highest-selling item category, followed by Snack Foods.
- Tier 3 has the highest total sales, while Tier 2 has the highest average sales per item.
- Supermarket Type3 has the highest average item sales among the outlet-type combinations shown.
- Outlet_Type is the strongest feature in the XGBRF model.
- Store_age, Item_MRP and Outlet_Identifier are the next most influential predictors.
- The final XGBRF model reaches a test R^2 of 0.6174 with MAE of 714.73 and RMSE of 1,019.73.

11. Business Interpretation

- Outlet format appears to be a major driver of sales, so outlet-specific strategies may be more effective than a single nationwide approach.
- High-performing product categories such as Fruits and Vegetables and Snack Foods deserve continued availability and merchandising attention.
- Regional differences indicate that location tier should be considered when allocating inventory and setting sales expectations.
- The importance of Item_MRP suggests that pricing/value positioning is relevant to sales prediction.
- The predictive model can be used as a baseline for sales planning, although additional validation and tuning would be appropriate before operational deployment.

12. Limitations and Reproducibility Notes

- The report is based strictly on the supplied notebook and its recorded outputs.
- The notebook's file path is named city_day.csv, while the loaded schema and analysis correspond to the Big Mart Sales dataset. The file naming/path should be corrected before publishing a reproducible repository.
- The dataset itself was not supplied as a separate file with the notebook, so the exact raw-data file should be added to the project or its source documented for full reproducibility.
- Ordinal encoding is used for categorical variables; alternative encodings could be evaluated.

- The final model uses default XGBRF settings in the notebook; systematic hyperparameter tuning could potentially improve performance.
- The model evaluation uses a single 80/20 train-test split for the final metrics, while cross-validation is used for model comparison.

13. Conclusion

The project demonstrates a complete retail analytics workflow from data quality assessment and missing-value treatment to exploratory analysis and predictive modelling. The analysis identifies meaningful differences by outlet, product category and location tier. The modelling results show that outlet characteristics and product pricing-related variables are especially informative for predicting Item_Outlet_Sales. The XGBRF model provides a solid baseline, achieving a test R^2 of 0.6174 under the notebook's current methodology.

14. Tools and Technologies

- Python
- Pandas and NumPy
- Matplotlib and Seaborn
- Scikit-learn
- PyMC and ArviZ
- XGBoost
- Google Colab

15. Author

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Project focus: Retail Sales Analytics, Exploratory Data Analysis and Machine Learning